Problem Statement:

Abstract:

News image caption generation is a specialized task that extends beyond conventional image captioning by requiring the integration of both visual data and the textual context of a news article. This task involves generating captions that not only describe the visual content of the image but also incorporate relevant named entities and information from the associated article. Unlike generic image captioning, which relies solely on visual features, news articles provide a much broader context, containing background information, named entities, and relationships that are crucial for producing meaningful and accurate captions. Traditional image captioning models often struggle with these contextual demands, leading to incomplete or inaccurate captions. Recent advancements, including Transformer-based models and commonsense knowledge reasoning, have shown promise in addressing these challenges by effectively leveraging both visual and textual inputs. This paper reviews state-of-the-art methods, highlights challenges such as named entity recognition and multimodal context integration, and proposes potential improvements to align generated captions with both the visual and narrative content of news stories.

Problem Statement:

Generating captions for news images presents unique challenges that go beyond those of conventional image captioning. While traditional models focus solely on visual features, they often miss the broader context provided by the accompanying article, which is crucial for generating accurate and informative captions. The article typically offers additional background, named entities, and relationships that cannot be inferred from the image alone. Current datasets and models are limited in their ability to incorporate this context, resulting in captions that are either incomplete or irrelevant. Furthermore, the presence of named entities—such as people, places, and events—adds complexity, as these entities are often absent from standard training datasets, making them difficult to predict. Addressing these issues requires models that can effectively integrate both visual content and textual context, enabling them to generate captions that are not only visually descriptive but also contextually accurate and informative.